

Neuronal diversity in the retina

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The listing of cell types present in the retina is nearing completion, the first time this can be said for any significantly complex sample of the central nervous system. Mammalian retinas contain ~55 separate neuronal types. The functions of 22 of them are known or can be strongly inferred. For these 22, in every instance, a cell defined as a 'type' by structural criteria carries out a distinct and individual physiological function. Electrophysiological experiments continue to reveal new features of the retina's handling of information, and there is every reason to believe that the remaining 33 types of cell will also have distinct physiological functions. Further subtleties clearly exist in both peripheral and central visual coding.

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Introduction

In trying to understand the working of a neural structure, or any other machine, an early goal is to catalogue the gears and wheels that make the machine run. For the central nervous system, that is more easily said than done. A few cells express cytochemically distinctive markers, but cell-type-specific proteins have yet to be found for most neurons [1]. In the retina, alternative methods, some of them quantitative, have made such a listing possible. Here, I give an overview of the cells present in mammalian retinas and present evidence that cell types defined by anatomical criteria represent distinct physiological entities.

Some important and lively areas are skipped: the controversy over the anatomical basis of trichromatic vision [2,3]; advances in the cataloguing and localization of neurotransmitter receptors [4,5•]; and the debate over presynaptic versus postsynaptic computation of direction selectivity [6•,7•]. Studies of coding by ganglion cell spike trains were recently reviewed [8,9]. Rather than any specific mechanistic issue, this review focuses on the overall, global view.

Mammalian retinas contain ~55 morphological types of neurons

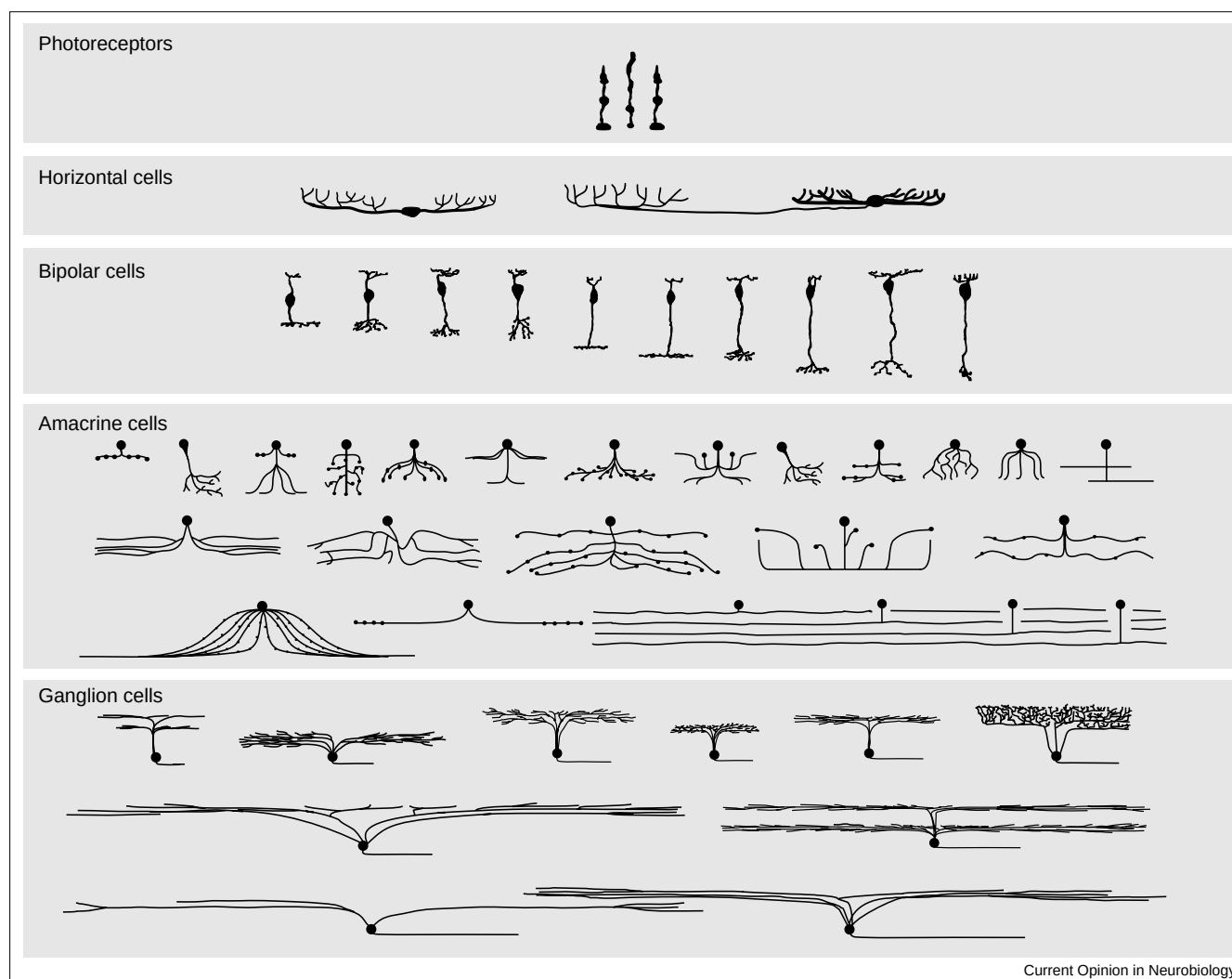
The major cell types present in a typical mammalian retina are illustrated in Figure 1. All except the bipolar cells, which are from the rat [10], are taken from studies on the rabbit, primarily because the most diverse type, amacrine cells, have been studied most systematically in the rabbit. There is much information on other mammalian species and general agreement that most of the specific cell types are conserved [11–15].

How were different cell types defined? Although some cells can be stained as populations by cytochemical markers, the mainstay remains the shapes and arborizations of the cells. An insight bequeathed by Cajal was that shapes of retinal neurons define their connectivity. The retina is a laminated structure, divided not only into five major layers but sublayers within them. Thus, Cajal's division of the inner plexiform layer into substrata was not merely a taxonomic convenience; it is a guide to the cells' levels of arborization, which in turn are the visible expression of their synaptic connections. For example, it is a physical impossibility for a bipolar cell whose axon is confined to layer 1 of the inner plexiform layer to make direct synaptic input onto a ganglion cell whose dendrites are confined to layer 5. The lamination of the inner plexiform layer thus serves as an initial wiring diagram of the inner retina and it was on this basis that Cajal sorted retinal neurons, revealed by the Golgi method, into distinct functional types. Modern techniques, to be discussed below, are less capricious than the Golgi, but the fundamental way of distinguishing one cell type from others remains the same.

How complete is this inventory?

How do we know that the catalogue is nearly complete? Photoreceptors and horizontal cells can easily be stained for specific proteins and/or distinguished by classical cytology. The bipolar cells shown in Figure 1 were identified by two different methods [10]. Approximately 60% of all bipolar cells were revealed by immunocytochemical markers. The remainder were identified by microinjection with Lucifer Yellow. When no new cells appeared after a large series of stainings and injections, the population was assumed to be complete. This can also be compared with previous studies of the monkey and cat retinas which reveal — with the exception of the midget cell — a similar array of bipolar cell types [11,16,17].

For amacrine cells, a systematic, unbiased sampling method was used [18,19•]. Photofilling is a photochemical technique in which a fluorescent product is created within an individual cell by focal irradiation of its nucleus. The photofilling method is quantitative, in the sense that 94% of targeted cells are successfully filled. This allows a systematic sampling method in which cells are chosen strictly at random for filling. A population of 261 neurons was targeted, allowing an estimate of the frequency of each type. Each of these cells was imaged digitally in three dimensions, and a classification of the cells was made. Subsequently, the digital archive was compared with a series of 49 retinas stained as wholemounts by the Golgi method. The two methods were found to confirm one another. Although the classification of a few cells was refined and two new types were identified, most of the cells classified by the photofilling method appeared similar

Figure 1

The major neuronal populations of a generic mammalian retina. For steric reasons, only a subset of the wide-field amacrine and ganglion cells are shown; the total number of cells in the illustration is 49. The illustration is generic in the sense, discussed in the text, that these cell populations seem to be present in all mammalian retinas. Known differences in primates are: a third type of cone, a midget bipolar cell, and a midget ganglion cell. In the primate fovea, these specialized cells are numerically dominant. However, the fovea represents a small fraction of the total primate retina and most

current studies of the central visual system in primates do not study the fovea; for practical reasons the usual window lies five to ten degrees peripheral to the foveal center. A point worth noting is that the numbers of cells are distributed rather uniformly among the different classes, except in the fovea. Wide-field cells are less numerous than narrow-field cells, as would be expected for reasons of coverage, but otherwise there are no predominant types. In other words, the retina is not composed of a few major players, but instead a large cast of more or less equipotent ones.

when seen by the Golgi method. A total of 29 types of amacrine cell were distinguished.

Three laboratories have undertaken systematic surveys of ganglion cells: Dacey's in the monkey, Berson's in the cat, and my own in the rabbit. The studies of cat and monkey primarily use visually-guided microinjection [20,21]. This method is less capricious than the Golgi, because it targets cells on the basis of size and/or staining of the soma with vital dyes. It also allows, in the more recent work, physiological identification of the cells. The most complete studies thus far are those of Berson and colleagues, who in

a series of meticulously detailed papers have described 10 types of retinal ganglion cell in the cat [22,23,24^{*}]. Less has been published from the monkey [14,25^{*},26,27], but its diversity is clearly similar to that of the cat. In the rabbit we are seeking completeness by using four mechanistically independent methods for filling cells: photofilling, microinjection, particle-mediated introduction of the gene for green fluorescent protein, and particle-mediated transfer of diI [28]. At this point we have filled 669 cells and are no longer seeing new cells. The ganglion cells shown in Figure 1 come from this series. Further fine-tuning will doubtless occur, particularly for the wide-field amacrine

and ganglion cells: because of their wide dendritic spread they can achieve a high retinal coverage with a small absolute number of cells. But the various methods, taken together, appear to provide a virtually complete picture of the major types of cells present in mammalian retinas.

Is the division into types overly fine?

How can one be sure that too many types have not been created? This is the old argument between ‘lumpers’ and ‘splitters’: the issue is whether, in obsessing over details, anatomists make mountains out of molehills, creating ‘types’ of cells that are in truth only variants of a single class. One of the reasons to think this has not occurred has already been mentioned: the stratification of cells in the retina puts rigid physical limits on the contacts that these can have, so that cells participating in different sublaminae of the inner plexiform layer are necessarily connected to different sets of presynaptic and postsynaptic partners. A second reason is that a sizeable fraction of cells can be distinguished morphologically and immunocytochemically [4,5,13,29]. As different neurotransmitters, receptors, and intracellular signaling molecules have different physiological consequences, the fact that cells of specific shapes are associated with specific constellations of proteins makes it undeniable that those cells have clear-cut functions within the retina’s circuitry.

A final method of validation is to verify that the number of cell types allows efficient coverage of the retina. This has been most systematically done for amacrine cells, where an estimate of the frequencies of the cells was available from the photofilled sample. Coverages were computed on the basis of observed frequency in the randomly photofilled sample and the known total number of amacrine cells. Coverages of all the narrow-field and medium-field cells hovered near one, the value that represents efficient tiling (wide-field amacrine cells overlap extensively and cannot be analyzed in this way but their absolute numbers, starburst cells excepted, are very small). If one assumes that most retinal neurons have a coverage of at least one, this provides insurance against the artificial creation of too many types: if a fixed total number of amacrine cells had been divided into too many types, there should have been amacrine cell ‘types’ with coverages less than one.

The structural type and physiological function of 22 types of retinal neuron are known: in every instance, a neuron independently defined as a structural type has a distinct physiological function

Because it has grown incrementally, the list of cells whose function are known may seem surprisingly long. Here is a summary overview, proceeding from outer to inner retina. Recent findings are referenced; the older evidence is discussed in the reviews cited elsewhere in this article.

Photoreceptor and horizontal cells

The structure and function of rods and cones are well known. Most mammalian retinas have two chromatic types

of cone, a few have three. The generic function of horizontal cells is to feed back to photoreceptor and/or bipolar cells — they subtract a pooled wide-field signal from the excitatory input. Horizontal cells come in two types, one of which possesses an axon with a finely-branched terminal feeding back onto rods. The likely reason for this specialization is that rods evolved long after cones [30,31], so that a simple evolutionary adaptation was to add an extra process onto an already existing cone-associated horizontal cell.

Bipolar cells

Rod bipolar cells receive input from rods and make synaptic output onto AII amacrine cells, which then synapse upon cone bipolar terminals and thence retinal ganglion cells. Once again, the apparent reason for this arrangement is the late evolution of rods. The rod pathway piggybacks on the pre-existing cone circuitry, so that the complex circuits of the inner retina did not have to be re-invented for the rod pathway.

A specialized blue cone bipolar cell selectively receives input from blue cones and transmits a chromatically-tuned signal to the inner retina. In primates, the midget bipolar cell provides a private line from individual cones to ganglion cells. A group of diffuse cone bipolar cells are stratified in different levels in the inner plexiform layer and are divided between ON and OFF types, excited by light onset or cessation, respectively. The CD-15 bipolar cell appears to make a selective synaptic output onto the OFF arbor of the ON-OFF direction-selective cells in the rabbit [32]. As will be discussed below, one function of the diffuse bipolar cells is to divide the output of cones into high frequency and low frequency domains.

Amacrine cells

Amacrine cell AII, as mentioned above, is a bridge between the rod pathway and the ganglion cells, and also serves to sharpen the temporal response of the rod system by a regenerative response to synaptic input. The indoleamine-accumulating amacrine cells (named A17 in the cat and rat [33]) support a reciprocal, feedback synapse upon the rod bipolar cells. They receive input on varicosities from rod bipolar cells and immediately feed back an inhibitory GABAergic signal. This causes the rod bipolar cell to be transient in its output, operating as a relatively high pass filter [34].

Starburst amacrine cells are cholinergic neurons that, because of their distribution of inputs and outputs, provide a feed-forward excitation onto certain ganglion cells. This causes the ganglion cells to be especially sensitive to moving stimuli. Starburst cells also participate in trans-retinal waves during early development [35–37]. Dopaminergic amacrine cells are sparse, wide-field cells that have resting spontaneous spiking modulated by light [38,39]. They control the sensitivity of many retinal neurons during light and dark adaptation. Amacrine cell AI is a spiking amacrine cell with a medium-field dendritic arbor and

wide-spreading, axon-like processes that traverse great distances through the retina. Because of their geometry, these are centrifugal cells, sending waves of inhibition across the retina in a radial pattern [40–42]. They may be involved in shift effects and long-range oscillations.

Ganglion cells

Ganglion cells in various species come in 10–15 different types. Most or all mammalian retinas contain a blue-ON ganglion cell, which receives the output from blue cone bipolar cells. A common feature of most mammalian retinas is the α -type ganglion cell, which corresponds to the brisk-transient or Y cell of physiologists. Similarly, many retinas contain a β -like ganglion cell corresponding to the brisk-sustained or X type [13]. In the monkey, a number of other morphologic variants have been identified but not yet characterized physiologically. In the rabbit, a bistratified ganglion cell is the ON-OFF directionally selective cell. A monostратified cell, sometimes termed δ -cell, is the ON-type direction-selective cell, which projects to the accessory optic system and drives the optokinetic nystagmus. A small, monostратified cell corresponds to the local edge detector of Levick and colleagues [43,44]. Levick [43] described several other physiological types of ganglion cell in the rabbit and cat, but these have yet to be securely mapped onto their anatomical counterparts.

Subtle structure underlies subtle function

Although much beloved by students of neuronal structure, the anatomical details of retinal cells, beautiful as they are, are not important here. The more general point is that cells defined as specific entities by morphological criteria have thus far, without exception, turned out to have distinct physiological functions. This implies a level of complexity of neural processing beyond the usual textbook description.

Modern methods of stimulation, recording and analysis show computations commensurate with the retina's anatomical complexity. An example is the predictive response of retinal ganglion cells to moving stimuli, in which the peak response precedes movement across the center of its receptive field [45•]. Another is the apparent multiplicity of controls of contrast sensitivity, ranging from the classic contrast gain control [46,47] to a slower, wide-field 'contrast adaptation' [48] to a highly localised adaptation that can occur in subregions within a ganglion cell's receptive field [49•]. Diffuse bipolar cells have now been directly demonstrated to divide the temporal frequency domain into separate high and low pass channels [50•,51•,52•]. Finally, the receptive field profiles of many ganglion cells are much less regular than the classic Gaussian [53,54•]. This puts limits on their ability to signal the position of stimuli to central structures, which should be taken into account in models of visual performance.

Anatomy also points to some areas of ignorance. What is the physiological microstructure of the receptive field center? Many types of narrow-field amacrine cells tile the retina within the dendritic field, and thus the receptive

field center, of most ganglion cells [18,19]. If nothing else, this invalidates the textbook generalization that the function of amacrine cells is to carry information laterally across the retina — these narrow-field amacrine cells introduce little more lateral conduction than do bipolar cells. The existence of physiological subunits within Y cell fields has long been known. Are there several kinds of subunit? What are their functions? Why, on the other hand, are there so many kinds of wide-field amacrine cells? Surely six to eight specialized cells are not required merely to cause the shift effects [55]. Finally, what are the remaining physiological types of retinal ganglion cell, and how do they contribute to vision? It now seems quite possible that the geniculostriate system of most mammals receives input from more than the M and P, X and Y cells. What are those other inputs, and how do they contribute to central visual function?

Conclusions

The listing of cell types distantly resembles the listing of genes in the genome; now that we have all the players, the task is to find out what they do. This is being revealed by new types of physiological experiments, notably those using modern paradigms of stimulation and analysis. A task for the future is to incorporate these revisions of our concept of the retina into our thinking of the function of the central visual system. The retina of 1980, containing only X and Y cells, now seems a stick figure. It is clear that a retina that possesses 29 types of amacrine cells has still more tricks up its sleeve.

Acknowledgements

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